Forecasting Marine Wave Patterns using Deep Learning Techniques with Wave Monitoring Data

School of Computer Science and Engineering – Winter Semester 2025 - 26

Introduction

Oceanographic forecasting predicts the development of the sea state to assist in maritime navigation and coastal protection. DL models, **LSTM**, **GRU** and **TFT** are implemented for forecasting wave parameters and the model with best performance is the chosen which turns out to be TFT because of its attention head mechanisms and Variable Selection Network.

Motivation

Existing ocean wave prediction models are based on ocean physics, which need high computational cost and these techniques are struggling with real-time prediction. Classical ML models also failed to catch the temporal dependencies.

Scope of the Project

The analysis uses a dataset of around 43,454 entries. The dataset has a preprocessing stage to counter missing data from sensors. The result of the models are calculated to predict a vector of six parameters: Significant Wave Height (Hs), MaximumWave Height (Hmax), Zero Upcrossing Wave Period (Tz), Peak Energy Wave Period(Tp), Peak Direction, and Sea Surface Temperature (SST). The project includes the essential development of three different deep learning models namely LSTM, GRU and TFT. The examination is based on RMSE,MAE, MAPE, and R^2 values to identify the best-performing model for real world use.

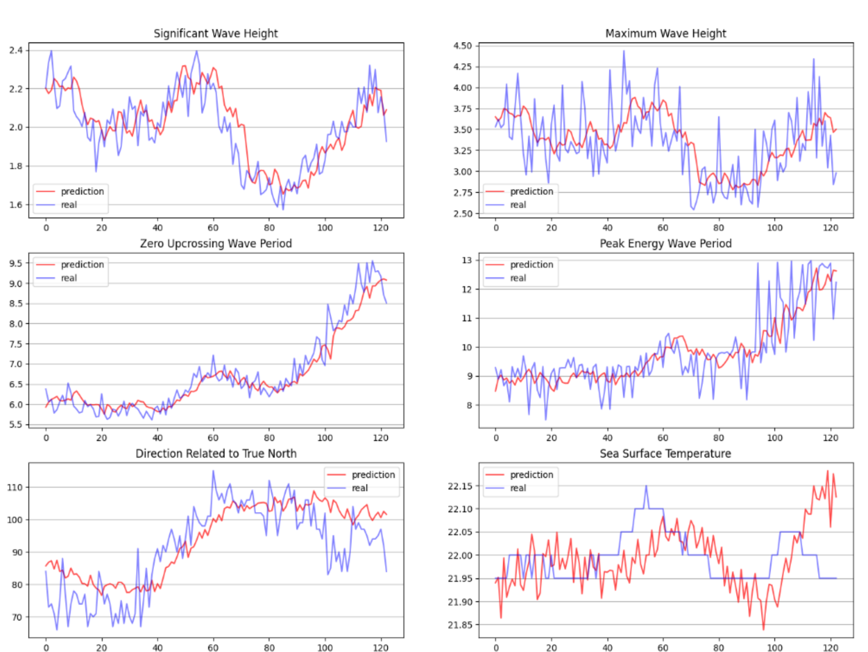
Methodology

In **preprocessing**, missing values were addressed and feature scaling was done.

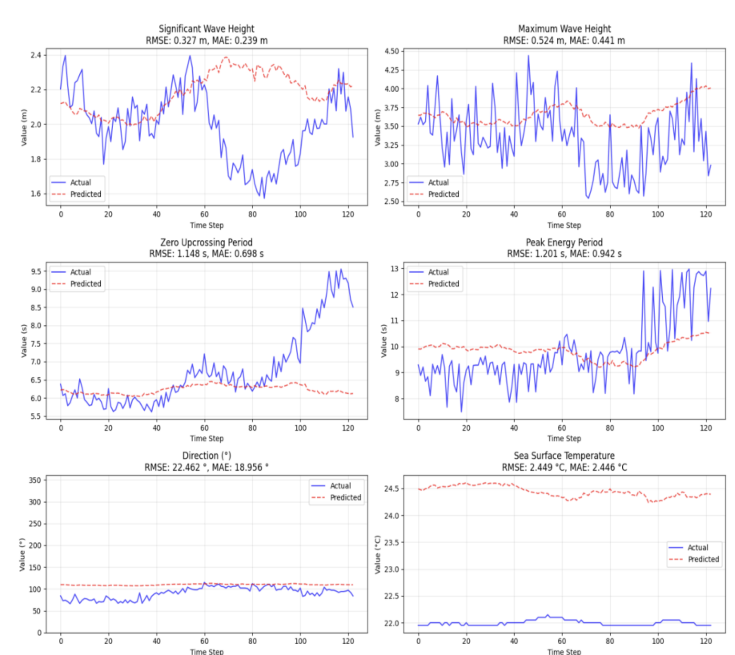
Dataset is partitioned into train and test sets with 42000 entries to train and the rest for testing.

LSTM:

- It reduces the vanishing gradient problem, very common in RNNs.
- The model contains three consecutive LSTM layers with 32, 16, and 10 hidden units, respectively.
- An integration of a dropout layer having rate of **0.2** between second and third layers of LSTM was done to reduce the overfitting risk.
- Adam optimizer was employed with learning rate = **0.001**



LSTM visualization of the Real and Predicted Values



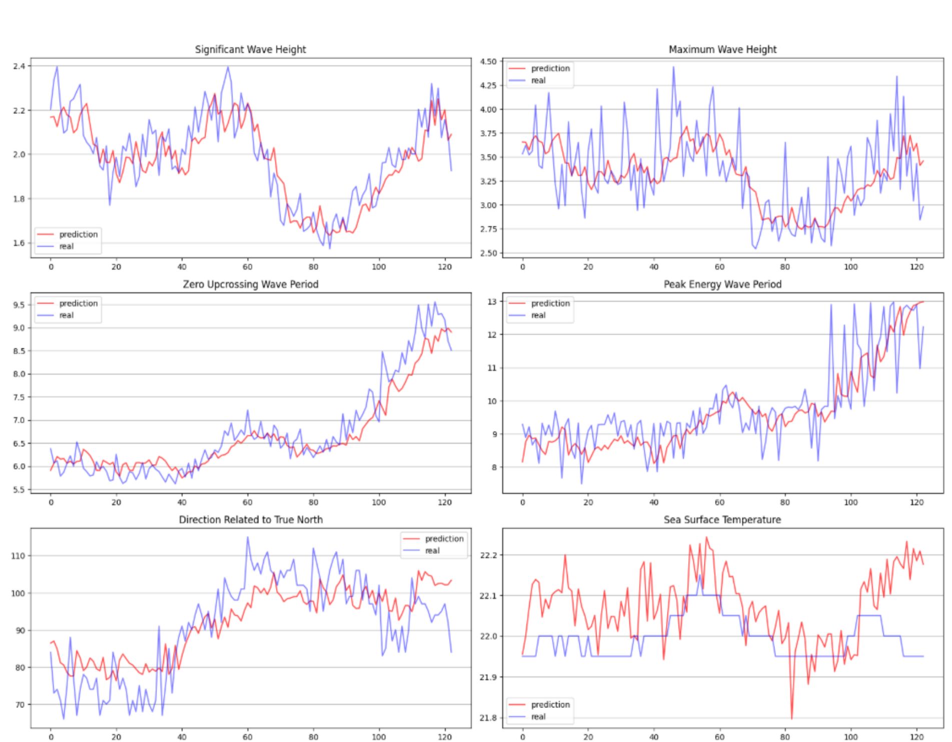
GRU visualization of the Real and Predicted Values

GRU:

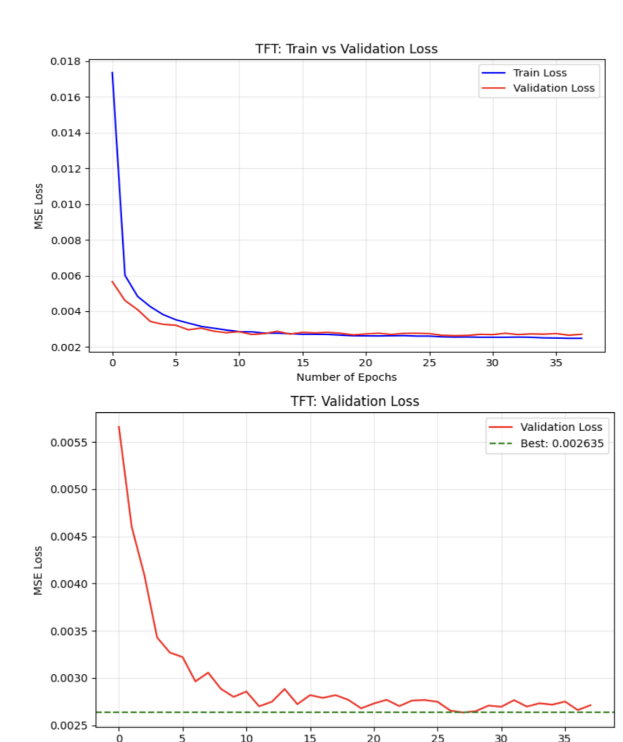
- The proposed GRU architecture consists of a deep stacked configuration of (128 to 64 to 32) is for hierarchical feature extraction.
- Elastic net regularization was applied where λ1–λ2=**0.001** on both kernel and recurrent weights to avoid overfitting.
- Progressive dropout rates (30%, 30%, 20%, 20%) has been applied to improve generalization and prevention of co-adaptation of neurons.
- Adam optimizer was employed with learning rate = **0.0005**, clipnorm = **1.0**

TFT:

- The TFT model was trained using processed multivariate time-series data for 50 epochs with Mean Squared Error (MSE) as the loss function and Adam optimizer (learning rate = **0.001**).
- AReduceLROnPlateauscheduler was applied to lessen the learning rate based on validation loss, and gradient clip ping was carried out to stabilize training.
- Model performance was monitored on the validation set with early stopping.



TFT visualization of the Real and Predicted Values



TFT learning curves

Results

The parameters namely RMSE, MAE, MAPE, and R^2 was used to evaluate performance across six target variables.

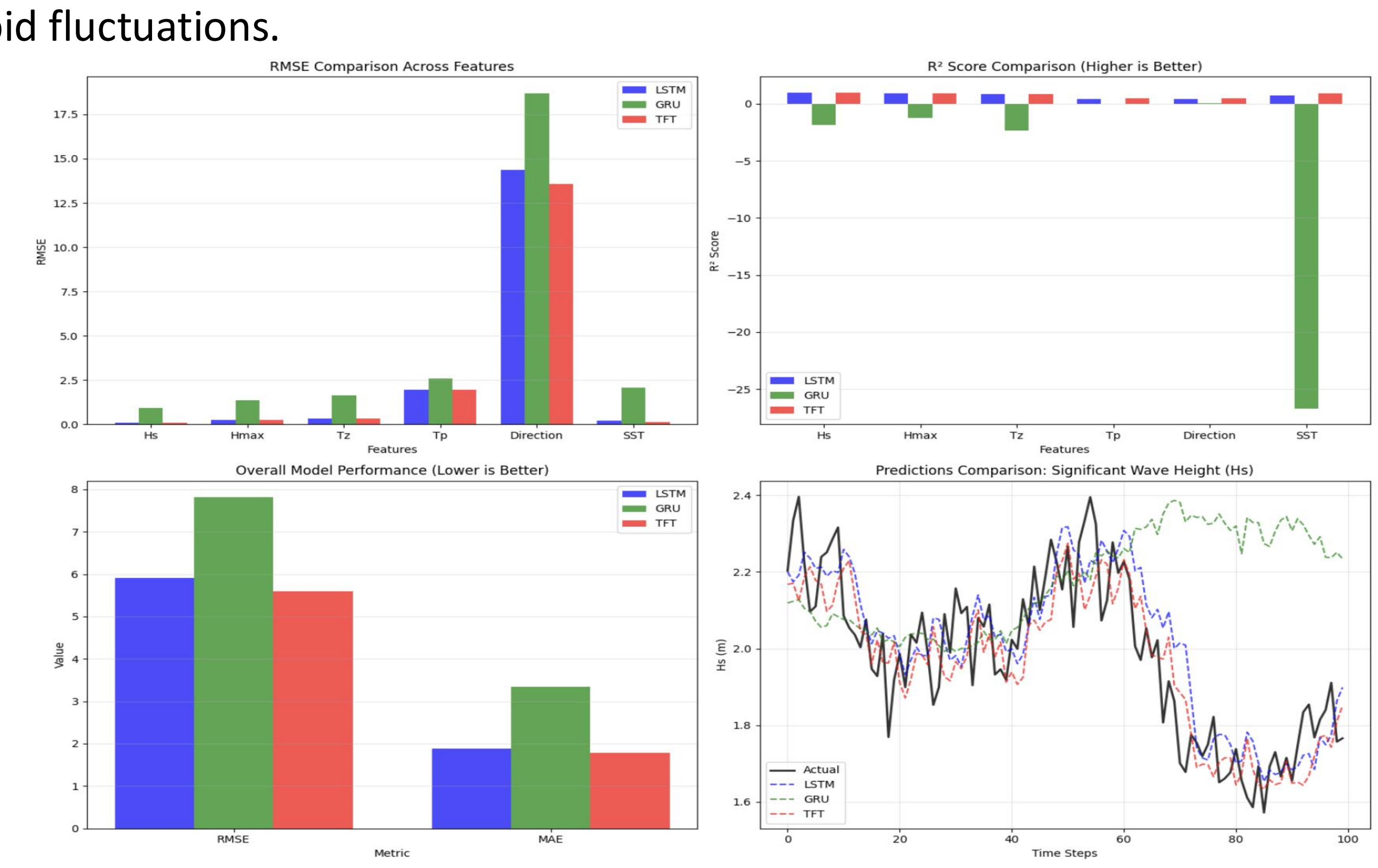
In RMSE analysis, TFT has resulted in better performance than both LSTM and GRU across all six parameters.

The MAE analysis was also performed across all three models. Out of 6, in 5 parameters, TFT achieved better performance than others. Only for Peak Energy Wave Period (Tp), LSTM marginally outperformed TFT.

MAPE calculates the average error in percentage which is helpful to interpret across different scale, achieved a similar result to MAE with TFT overall the best performing model.

R^2 score of TFT was calculated as 0.9797, which was the highest, followed by LSTM (0.9774) and GRU (0.9606).

Graphical representation also reveals that while LSTM and TFT prediction paths stay closer to the actual track, GRU exhibits larger deviation, particularly during rapid fluctuations.



Model Performance Comparison: LSTM vs GRU vs TFT

Model Comparison with RMSE

Feature	LSTM	GRU	TFT	Best Model
Hs	0.0889	0.9093	0.0824	TFT
Hmax	0.2484	1.3672	0.242	TFT
Tz	0.3416	1.6312	0.3194	TFT
Tp	1.9561	2.5959	1.9391	TFT
Direction	14.3529	18.6922	13.5743	TFT
SST	0.2117	2.0568	0.1148	TFT
Overall	5.917	7.8073	5.6006	TFT

Model Comparison with MAE

Feature	LSTM	GRU	TFT	Best Model
Hs	0.0673	0.8256	0.0611	TFT
Hmax	0.182	1.238	0.1723	TFT
Tz	0.2595	1.4075	0.2365	TFT
Tp	1.4329	2.1708	1.4345	LSTM
Direction	9.1924	12.4436	8.7315	TFT
SST	0.1528	1.9832	0.0893	TFT
Overall	1.8812	3.3448	1.7875	TFT

Conclusion

This research work has provided a detailed comparative analysis of three deep learning models: LSTM, GRU, TFT. TFT achieves its better performance from its attention-based architecture. Though **TFT** provides better performance, its computational complexity against deployment constraints should also be considered. In a situation where resources are limited, LSTM may give a balanced result of efficiency and accuracy.

References

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Demo Video Link:

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